OpenRocket Flight Simulation for a Hypothetical Planet (Mustafar)

This report explains the design process, flight performance, and environmental analysis of a two-stage sounding rocket designed in OpenRocket for the fictional planet Mustafar. The objective was to deliver a 1 kg scientific payload to an apogee of approximately 4000 m while maintaining stability, meeting recovery requirements, and remaining within all mission constraints.

1. Simulation Environment

To reproduce the Mustafar launch environment, the planetary parameters provided in the assignment were configured in OpenRocket. A custom planetary JSON file was imported to define the local gravitational acceleration of 6.5 m/s^2. Atmospheric conditions were then adjusted to match the mission requirements: pressure of 75 kPa, temperature of 700 K, air density of 0.374 kg/m^3, and wind speed of 10 m/s at a direction of 270°. These changes allowed the simulation to represent the reduced gravity and thin atmosphere of Mustafar.

2. Rocket Configuration

The rocket consists of three major sections: the payload section, sustainer stage, and booster stage.

Payload Section

The payload section contains an ogive nose cone, body tube, and the required 1 kg scientific payload. A drogue parachute is deployed at apogee to reduce descent speed immediately after the peak of the flight. A 165 cm main parachute is deployed later during descent to ensure a safe landing. This section contains no propulsion system.

Sustainer Stage

The sustainer uses a J415 motor and provides the primary high-altitude propulsion after stage separation. Five trapezoidal fins are installed on this section to improve aerodynamic stability. Initial simulations showed excessive stability margins of roughly 5 calibers. The fin configuration was therefore optimized to maintain a more practical stability range of approximately 1.5 to 2 calibers throughout the flight profile.

Booster Stage

The booster uses a J295-16 motor to provide initial launch thrust. Three trapezoidal fins are attached to maintain stability during the boost phase. Following burnout and separation, the booster recovers independently using a drogue and main parachute system.

3. Physical Characteristics

The entire rocket is constructed from cardboard, satisfying the mission material requirements. The total launch mass is 4.309 kg with motors installed and 2.033 kg without

motors. The overall rocket length is 147 cm, comfortably below the 220 cm mission limit. An ogive nose cone was selected to reduce aerodynamic drag and improve ascent efficiency.

4. Performance Analysis

The final configuration achieved an apogee close to the required 4000 m target while maintaining acceptable stability and recovery characteristics. The design emphasizes a balance between altitude, stability, mass, and recovery rather than simply maximizing altitude.

5. Earth vs Mustafar Comparison

The mission brief requires examination of how environmental parameters affect rocket performance. Several comparison simulations were therefore conducted.

Temperature and Pressure Effects

When Earth-standard atmospheric conditions were applied, the apogee decreased to approximately 1516 m. Earth's denser atmosphere generates significantly higher aerodynamic drag, which reduces the rocket's maximum altitude. This result demonstrates that atmospheric density is one of the most important factors influencing ascent performance.

Gravity Effects

Mustafar's gravitational acceleration is approximately 66% of Earth's gravity. When Earth gravity was used in the simulation, the apogee dropped to approximately 2125 m. The higher gravitational force increases vehicle weight and reduces the altitude gained from the same motor impulse. An additional recovery deployment warning was also observed during the Earth-gravity simulation.

Wind Effects

When the 10 m/s crosswind was removed, the rocket achieved an apogee of approximately 4613 m. The absence of lateral wind reduced aerodynamic losses and weathercocking effects, allowing the vehicle to climb more efficiently. This comparison illustrates the significant influence of wind on both altitude and flight path.

6. Design Decisions and Discussion

The rocket is relatively long compared to its diameter, making stability control particularly important. Additional fins were incorporated primarily to move the center of pressure to a suitable location relative to the center of gravity. Extensive simulation and iteration were required to keep the stability margin within an acceptable range while still meeting altitude requirements.

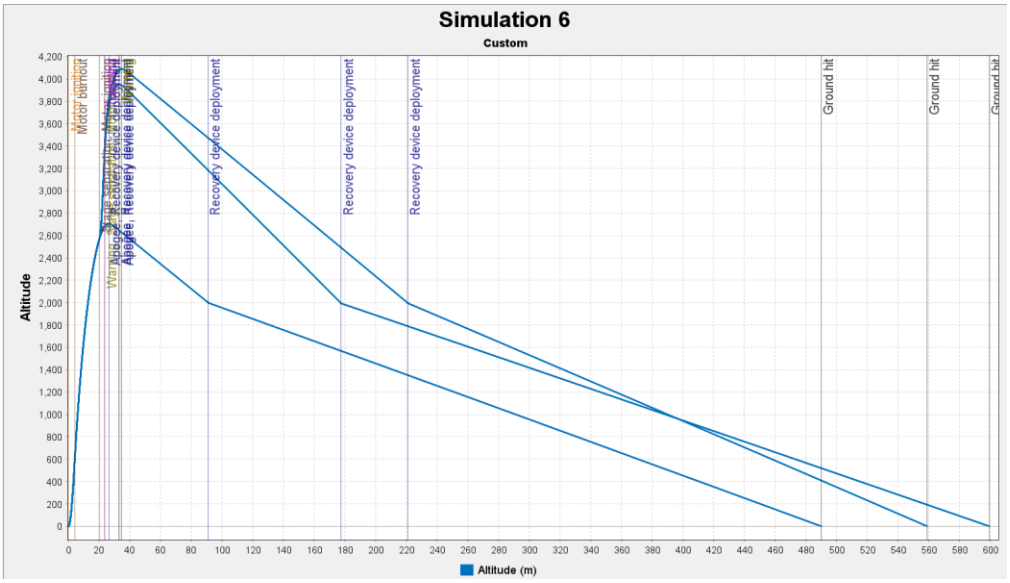
7. Conclusion

The final rocket successfully demonstrates the challenges of designing for an extraterrestrial environment. The Mustafar atmosphere and reduced gravity enable significantly higher performance than equivalent Earth conditions. Through careful

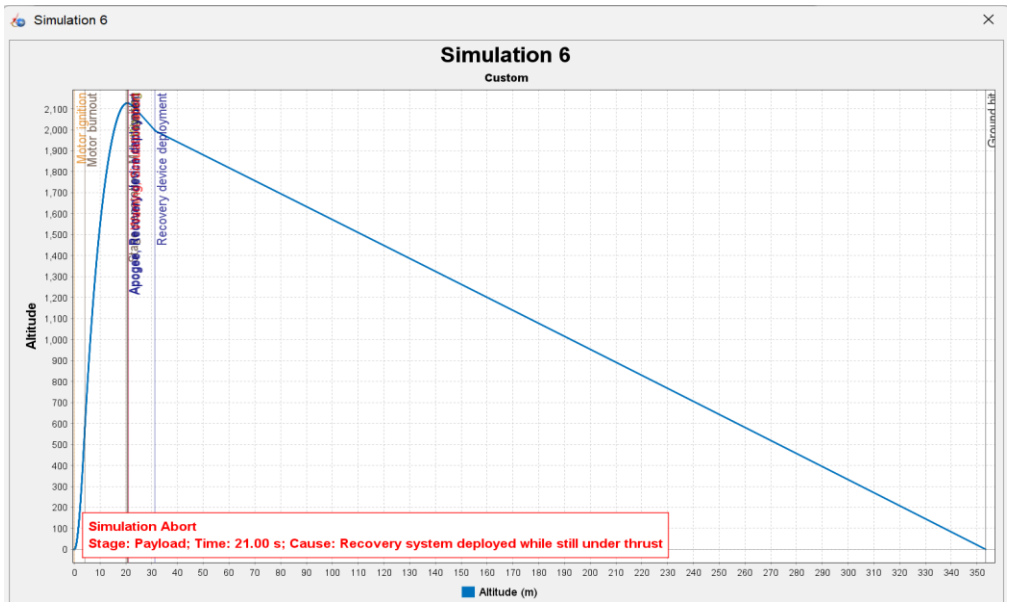
selection of motor combinations, fin geometry, structural configuration, and recovery systems, the rocket satisfies the mission objectives while remaining stable and recoverable throughout flight.

Here are the flight trajectories for different cases.

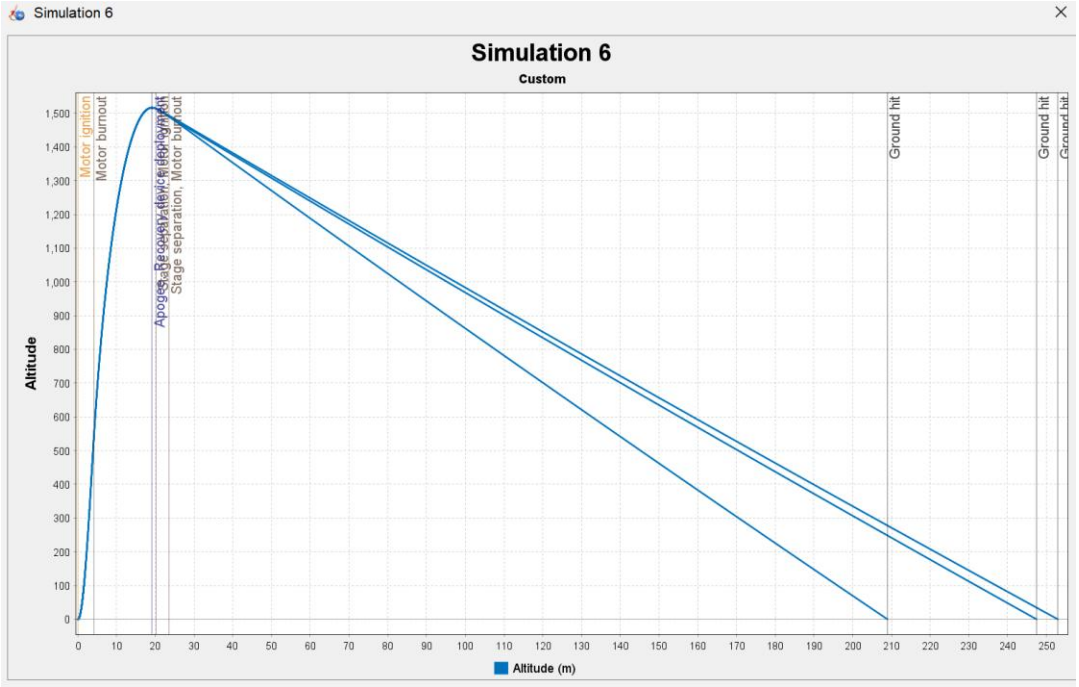
1) Normal Mustafar



2) Mustafar with Earth's Gravity



3) Mustafar with Earth's Temperature and Pressure



4) Mustafar without Wind

